

INDUSTRIAL INTERNSHIP REPORT
ON
“INTERNET OF THINGS”

Submitted in partial fulfillment of the requirement for degree of

B.CA
in
RV HIGHER EDUCATION TECHNICAL INSTITUTE
at
CHAUDHARY CHARAN SINGH UNIVERSITY

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I express my deepest thanks to **UP SKILL CAMPUS** for taking part in useful decision & giving necessary advice and guidance and arranged all facilities to make life easier. I choose this moment to acknowledge his contribution gratefully.

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ABSTRACT

Internet of Things (IoT) conceptualizes the idea of remotely connecting and monitoring real world objects (things) through the Internet [1]. When it comes to our house, this concept can be aptly incorporated to make it smarter, safer and automated. This IoT project focuses on building a “SMART HOME AUTOMATION SYSTEM” which is mainly concerned to design a system that allows users, upon authentication, to remotely control and monitor multiple home appliances using a cell-phone based interface. The leverage obtained by preferring this system over the similar kinds of existing systems is that the alerts and the status sent by the wifi connected microcontroller managed system can be received by the user on his phone from any distance irrespective of whether his mobile phone is connected to the internet. The microcontroller used in the current prototype is the TI- CC3200 Launchpad board which comes with an embedded micro-controller and an on-board Wi-Fi shield making use of which all the electrical appliances inside the home can be controlled and managed.

Internet of things has been governing the electronics era with cloud services dominating the ever increasing electronics product segment. By using this we can switch on or off the lights or fans automatically when we are outside. This proposes a smart home automation system based on Open source cloud server “things speak.com” and a low cost esp8266 Wi-Fi module.

The System transmits an alert signal to the Open source cloud which provides a alert signal on the users mobile phone. The system employs a second esp8266 module which is programmed to act as a web server, which allows the user to activate or deactivate the security system by means of any device with internet.

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1. INTRODTION TO INTERNET OF THINGS (IOT)

Anyone who says that the Internet has fundamentally changed society may be right, but at the same time, the greatest transformation actually still lies ahead of us. Several new technologies are now converging in a way that means the Internet is on the brink of a substantial expansion as objects large and small get connected and assume their own webidentity.



Following on from the Internet of computers, when our servers and personal computers were connected to a global network, and the Internet of mobile telephones, when it was the turn of telephones and other mobile units, the next phase of development is the Internet of things, when more or less anything will be connected and managed in the virtual world. This revolution will be the Net's largest enlargement ever and will have sweeping effects on every industry — and all of our every day lives.



Smart connectivity with existing networks and context-aware computation using network resources is an indispensable part of IoT. With the growing presence of Wi-Fi and 4G-LTE wireless Internet access, the evolution towards ubiquitous information and communication networks is already evident. However, for the Internet of Things vision to successfully emerge, the computing paradigm will need to go beyond traditional mobile computing scenarios that use smart phones and portables, and evolve into connecting everyday existing objects and embedding intelligence into our environment. For technology to disappear from the consciousness of the user, the Internet of Things demands: a shared understanding of the situation of its users and their appliances, software architectures and pervasive communication networks to process and convey the contextual information to where it is relevant, and the analytics tools in the Internet of Things that aim for autonomous and smart behavior. With these three fundamental grounds in place, smart connectivity and context-aware computation can be accomplished.

THE INDUSTRIAL INTERNET OF THINGS

The industrial internet of things (IIOT) refers to the extension and use of the internet of things (IOT) in industrial sectors and applications. The IIOT encompasses industrial applications, including robotics, medical devices, and software-defined production processes.



2. HISTORY

In early 1982 the concept of the network of smart devices was discussed, with a modified Coke machine. This coke machine is modified at —Carnegie Mellon University—and becoming the first Internet-connected appliance. This machine was able to report its inventory and whether newly loaded drinks were cold.

In 1994 Reza Raji explained the idea of IoT as “small packets of data to a large set of nodes, so as to integrate and automate everything from home appliances to entire factories”. After that many companies proposed various solutions like Microsoft’s at Work or Novell’s Nest. Bill Joy proposed Device to Device (D2D) communication as a part of his —Six Websl frameworks at the World Economic Forum at Davos in 1999.

The thought of Internet of Things first became popular in 1999. British entrepreneur Kevin Ashton first used the term Internet of Things in 1999 while working at Auto-ID labs. Besides that near field communication, barcode scanners, QR code scanners and digital watermarking are the various devices which are working on IoT in the present scenario.

3. DEFINITION

—Today computers and the Internet are almost wholly dependent on human beings for information. Nearly all of the roughly 50 petabytes (1 petabyte=10¹⁵ bytes) of data available on the Internet were first captured and created by human beings by typing, pressing a record button, taking a digital picture, or scanning a bar code. Conventional diagrams of the Internet leave out the most numerous and important routers of all - people. The problem is, people have limited time, attention and accuracy all of which means they are not very good at capturing data about things in the real world. And that's a big deal. We're physical, and so is our environment ... You can't eat bits, burn them to stay warm or put them in your gas tank. Ideas and information are important, but things matter much more. Yet today's information technology is so dependent on data originated by people that our computers know more about ideas than things. If we had computers that knew everything there was to know about things using data they gathered without any help from us we would be able to track and count everything, and greatly reduce waste, loss and cost. We would know when things needed replacing, repairing or recalling, and whether they were fresh or past their best. The Internet of Things has the potential to change the world, just as the Internet did or even more.

4. ARCHITECTURE OF INTERNET OF THINGS

Architecture of internet Of Things contains basically 4 layers:

- Application Layer
- Gateway and the network layer
- Management Service layer
- Sensor layer

APPLICATION LAYER:

- Lowest Abstraction Layer
- With sensors we are creating digital nervous system.
- Incorporated to measure physical quantities
- Interconnects the physical and digital world
- Collects and process the real time information

GATEWAY AND THE NETWORK LAYER:

- Robust and High performance network infrastructure
- Supports the communication requirements for latency, bandwidth or security
- Allows multiple organizations to share and use the same network independently

MANAGEMENT LAYER:

- Capturing of periodic sensory data
- Data Analytics (Extracts relevant information from massive amount of raw data)
- Streaming Analytics (Process real time data)
- Ensures security and privacy of data.

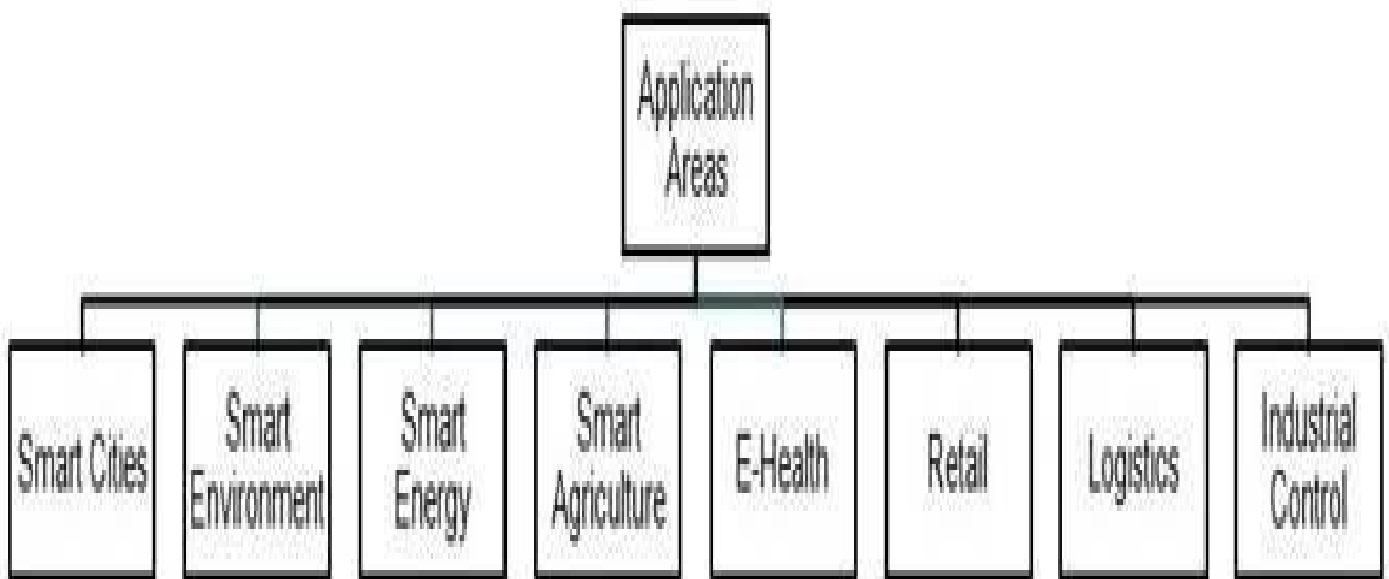
5. APPLICATIONS

There are several application domains which will be impacted by the emerging Internet of Things. The applications can be classified based on the type of network availability, coverage, scale, heterogeneity, repeatability, user involvement and impact.

We categorize the applications into four application domains:

- (1) Personal and Home
- (2) Enterprise
- (3) Utilities
- (4) Mobile

There is a huge crossover in applications and the use of data between domains. For instance, the Personal and Home IoT produces electricity usage data in the house and makes it available to the electricity (utility) company which can in turn optimize the supply and demand in the Utility IoT. The internet enables sharing of data between different service providers in a seamless manner creating multiple business opportunities.



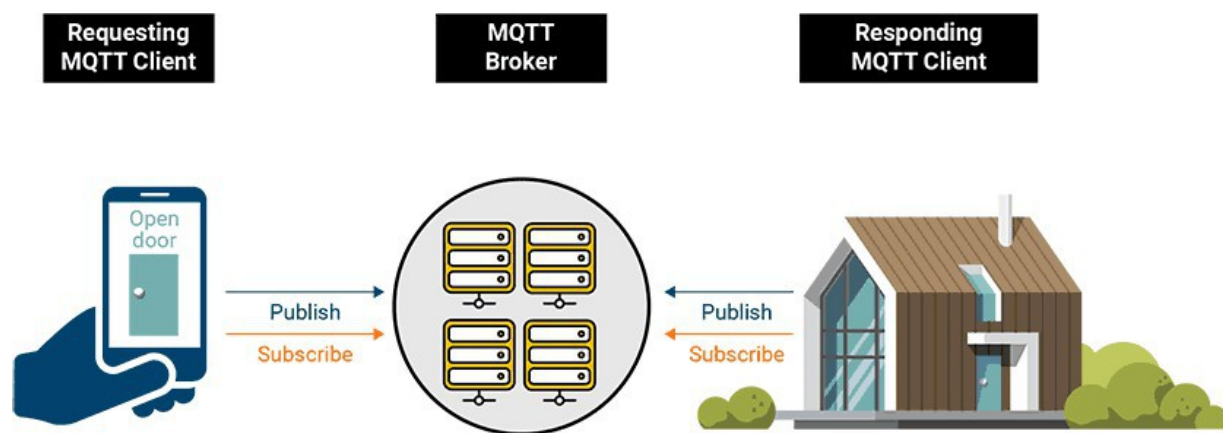
5.1 Home Automation using MQTT Protocol

Home automation is a new buzz now a days ..there is so many open source hardware available to build your own home automation. So many tools, its easy to confuse you to decide which is best /suitable for you. Keeping cost in mind and the basic requirement in mind you can find the best and cost effective solution in this instructable to get started with your first automation project.

Target:

Target of this instruct able is to control your home appliance by your android phone.

When the you pews the button in your phone light or fan in your home turns on or off.



MQTT protocol:

In a simple language when you want to communicate to your controller through mobile or a web browser you need a communication protocol and MQTT does that magic. MQTT is a message queuing telemetry transport. It is specifically designed for automation. So where does it do magic?? its a light weight protocol, what is light weight means?? Well it's having a very low footprint to send and receive data. So it uses very less amount of data to send and receive and mobile app using MQTT also consumes less battery.

So MQTT is the best for This project.

MQTT works on publish subscribe events. It requires MQTT broker in between. For got all these thing I explains MQTT with our example.

When you press the switch in your mobile or any web device you are telling your controller that you want to turn on the switch. Now as in MQTT protocol MQTT broker seats in between your mobile device and controller. Broker take care of successfully transfer your command to the controller.

5.2 EDUCATIONAL APPLICATION

Internet of Things (IoT) has impacted the educational institution strongly. Education institutions are opting for the IoT ecosystem and utilizing various techniques ranging from augmented reality to cloud computing. With the integration of IoT technology, physical environments are turning smarter and more interconnected than ever. When discussing IoT in education, one might imagine things like an advanced smart classroom, digital board, and voice-based command system.

However, IoT has progressed much further and can be utilized to keep the parents updated regarding the pickup and drop of their kids, smart security cameras in the school premises, automated tracking of student attendance and much more can be accomplished within the IoT ecosystem.



With the advancement in technology, our next generation is already addicted to the internet and information on demand. The major question arises, will IoT in the education system have a positive or negative effect on students. Many challenges such as privacy and additional cost of technology in education are currently being studied. Many educational institutes are still reluctant to accept IoT and are finding a better fit for it in the system. IoT in education can be broadly classified into student's and staff's perspectives.

From the staff perspective, IoT has the potential to manage the class attendance of the students and the availability of the necessary educational equipment. Installing the IoT sensors in multiple locations within the educational system such as the library, cafeteria, and entrance helps in tracking and monitoring the student's activity at different locations.

6. Applications of IoT in the Educational Sector

IoT plays a vital role in transforming the traditional educational system. There is a wide range of IoT applications in the educational sector. Let's see a few of the areas in the educational sector where IoT can be applied.

1. Energy Management

Internet of things can be utilized for energy management by installing IoT sensors in lights and water taps. IoT based organization claims IoT solutions can help in creating a green campus by cutting down CO₂ emission. IoT can help campus monitor their energy consumption and make sure energy is used efficiently with the help of the smart grid. With the IoT ecosystem, energy and water usage are controlled and monitored thus establishing a healthy teaching atmosphere.

2. Security and remote classroom Access

One of the major concerns of an educational system which is creating a safe and secure atmosphere for students can be achieved by opting for the IoT ecosystem. Embedded technologies such as NFC can be utilized to manage students and their access to various parts of the campus such as laboratories and other places within the educational institute. A real-time classroom control mechanism can be developed with the help of NFC .

3. Health monitoring of IoT in Education

With the help of wearable devices, student's health can be regularly monitored and can detect physiological signals over time. With the help of IoT, signs of depression and suicidal intentions can be detected in students and thus giving enough time for preventing any tragedy.

4. Educational Applications and Interactive learning

Educational apps facilitate customization and allow students to decide their course of action in the way subject knowledge is delivered. From home computers to televisions students can access educational live contents on multiple channels when on sick leave. Certain applications can be utilized to create a 3D graphical textbook that features video content along with notes. Taking notes is no longer writing down on a piece of paper, with the advancement in IoT students can read out loud and voice-based application converts the speech to text and saves them in the digital notebook.

7. ROLE OF IOT IN DIFFERENT SECTORS

1. AGRICULTURE APPLICATION

In the era of the internet and connected devices Internet of Things (IoT) is the next big thing for the industry. On the other hand, it is anticipated that by the next 30 years, the population of the world will exceed 6 billion, and the incremental throughput needed to produce food for this population is 70%. Incorporating IoT based smart agriculture systems is essential to cope up with this need. Let's take a look at the challenges facing the agriculture industry and how IoT is an answer to the problems.

Following are some of the important use cases of the IoT in the agriculture industry.

1. Real-time crop monitoring

Smart sensors, motion detectors, smart motion-sensing cameras, light detectors enable farmers to get the real-time data of their farms to monitor the quality of their products and optimize resource management.

2. IoT analytics in agriculture

Data from smart sensors can be analyzed for predictive analysis and automated decision-making. It helps farmers with smart, automated decision-making instead of classic rule-based systems or manual procedures. Predictive analysis and machine learning can help farmers to cope up with extreme weather conditions like floods, drought, etc. The main advantage of IoT analytics is the inclusion of soil quality, temperature, humidity as a parameter.

3. Livestock management

As industry agriculture strongly depends on the manual workforce. Given the changes in the global economic landscape, the supply-demand gap is increasing day by day. Livestock is one such area that needs regular monitoring. IoT based smart tracking can help farmers to get the stock information directly to their smart devices. It enables farmers with stock management to detect flue breakouts much earlier and separate infected breeds with non-infected ones.

4. Drone-based use cases

Industry grade drones also have multiple use cases in smart farming. On the one hand, drones are used to monitor air, soil, moisture quality; on the other hand; they can help with physical activities like automated spraying of fertilizers, preventing physical breakouts in farms, etc. Obviously, there are some challenges incorporating drones in farms, but successful deployment and use cases can help to reduce the manual workforce in agriculture significantly.

5. Smart greenhouse solutions

Greenhouses are conventionally used to maintain the necessary atmosphere for plants, and this process demands continuous monitoring and manual intervention. Industrial IoT solutions can be incorporated to automate this process. Data collected by smart sensors can be automatically analysed, and deep learning-based systems can be deployed to make decisions and create a certain climate automatically. With these smart sensors, climate variables and water consumption can be monitored via SMS or Wi-Fi-based systems.

6. Smart waste management

One of the main differences between agriculture and other industry is the value of industrial waste. Biological waste produced from farming can be reused for making fertilizers; IoT solutions can help manage the process remotely in a smart way. Smart sensors can be used to measure the presence of toxic chemicals in wastes and manage the proper agricultural lifecycle.

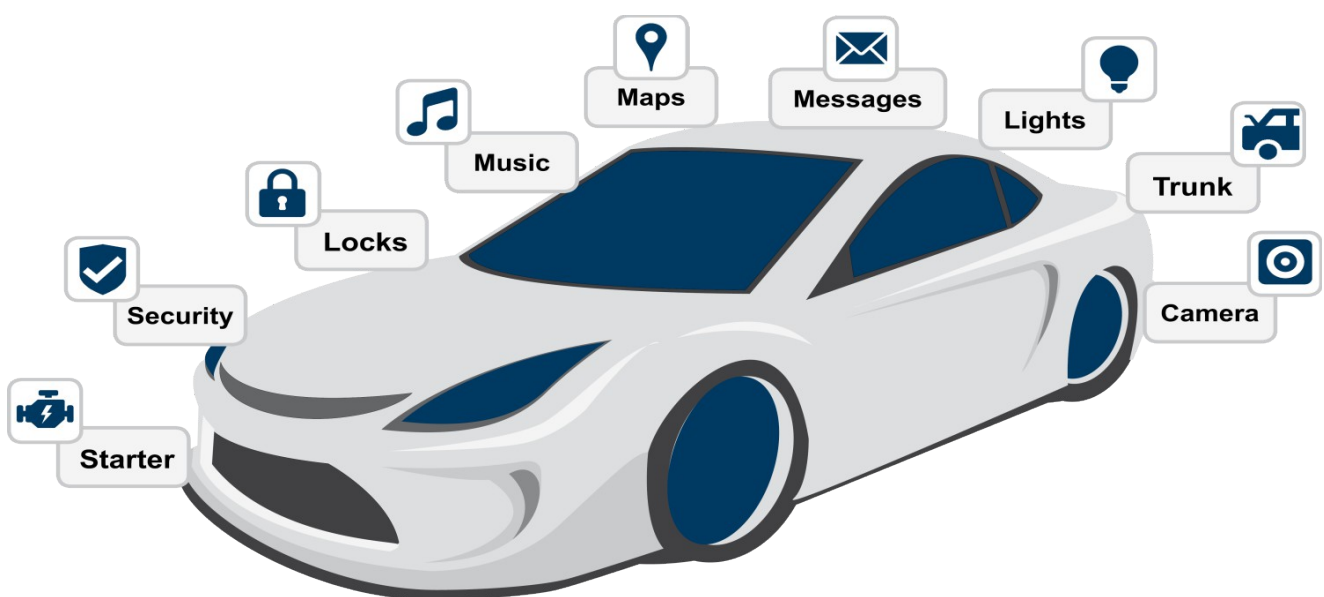


2. AUTOMOTIVE APPLICATION

IoT offers a wide range of innovative technologies for modern automobiles. It includes

- Wireless connectivity
- Advanced sensor networks
- Machine to machine communication
- Infotainment

All these innovations will make them more efficient than before.



3. IOT IN TRANSPORTATION

The transport sector plays a crucial role in the development process. Transportation needs depend on many factors such as goods availability, movement of travelers, logistics, etc. Transportation, therefore, becomes an important and integral part of connecting consumers to the supply chain team by logistics.

IoT integration helps in creating a centrally controlled network which can optimize the distance covered by the vehicle, find better and safe routes in case of a critical situation, efficiently manage and maintain the goods, material and purchase orders and overall gives a positive impact on the revenue generated by the transportation sector.

❖ **The Intelligent Transportation system will benefit four major segments:**

- **Travelers Segment:** People who will be benefited from the services provided by the Intelligent Transportation system.
- **Automobile Segment:** IoT will have a huge impact on the vehicles and will change the way how one interacts with the vehicle.
- **Infrastructure Segment:** Traffic management utilities such as traffic lights, cameras embedded with IoT will result in the smart management system.
- **Core Centers:** The data generated from the automobiles and infrastructure will be analyzed and communicated between different segments.

❖ **Applications of IoT in Transportation**

In the transportation and logistics business, the internet of things has numerous opportunities. Vehicles can be tracked for their speed, position, whether they are running or stopped, or at any danger, etc., using IoT. In most situations, trucks are used for the purpose of transportation or to carry heavy loads. Measuring the truck's interior conditions such as humidity, temperature, and the light condition is very critical during these periods. Using an IoT transport company can configure the payment service at the tolls or any parking tickets. IoT also allows vehicle guidance and navigation control systems (road, air, water transport) monitoring and controlling transport is highly possible with the use of IoT. Let's look at the different applications of IoT in transportation.

1. Transport Management System

Public Transport provides many advantages to city-wide travellers. But it was always a challenge to track the vehicle's real-time location and to know when the vehicle will arrive at a stop. Such problems were eradicated by Industrial IoT in Transportation and provided the Public Transporters real-time tracking solution.

2. Transportation Monitoring System

If the transportation company gets the vehicle's real-time facts and figures, they can make the most of the business benefits. IoT has steadily and slowly transformed the transport sector and the management of the fleet. The introduced technology improved the cost of maintenance, operating and fuel expenditure, and immediate response to incidents in real-time.

3. Asset Management and Remote Area Surveillance

The biggest challenge with assets that are being used in IoT is monitoring them remotely. Businesses go through a lot of difficulties to find the right assets and figuring how they can be used efficiently. IoT systems gather applicable information from the assets and carry out an insightful analysis of it.

Biz4Intellia, an end-to-end IoT software provider, helps you to monitor your fleet's real-time information such as loading weight on the truck, etc. In addition, the —Biz4Intellia IoT solution‖ can also be used to learn the latitude and longitude of the asset. The advanced analytics feature gives more information regarding geofencing, Propensity reading, and much more.

4. Inventory Management System

Smart Inventory Management serves as a stimulant for real-time information across the supply, processing and storage hub, thus decreasing inventory management costs. Such supply chains will combine large amounts of data to gain ways to create an integrated framework for inventory management.

Smart inventory management is based on supporting analytics. It has been known from recent studies that the potential contributions of IoT to Logistics, Inventory Management, and Supply Chain Management could range from \$570B to \$800B per year by 2025. The warehousing operations are accelerated by real-time inventory tracking with the help of IoT. Further, the sensors are used to gather accurate Inventory conditions.

5. Information and Entertainment in Automobiles

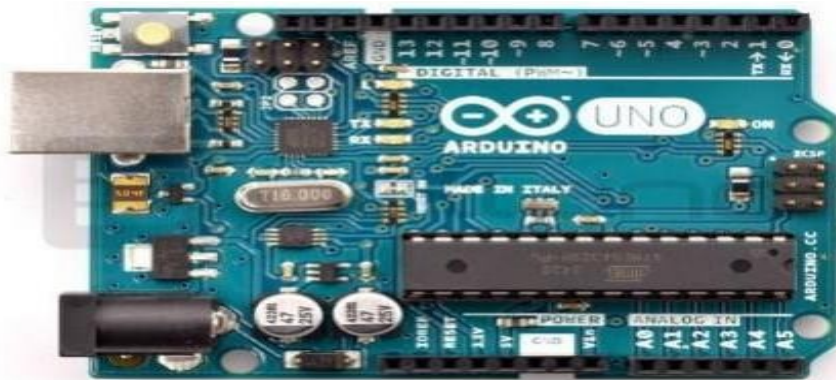
Wi-Fi-based applications have allowed vehicles to have entertainment and telematics functionality. By using automotive telematics, a car owner can have an eye on his vehicle even from a remote location. By means of a smartphone-enabled dashboard, car owners can always be assured of their safety. External sensors and cameras track the condition of the vehicles and send the data to the user on a mobile app.



8. Components Used

8.1 Arduino UNO

Arduino is an open source, PC paraphernalia and programming organization, endeavour, and client group that plans and produce microcontroller packs for constructing programmed devices and intelligent object that can detect and control questions in the real world. The inception of the Arduino extend began at the Interaction Design Institute in Ivrea, Italy. The equipment reference plans are appropriated under a Creative Commons Attribution Share.



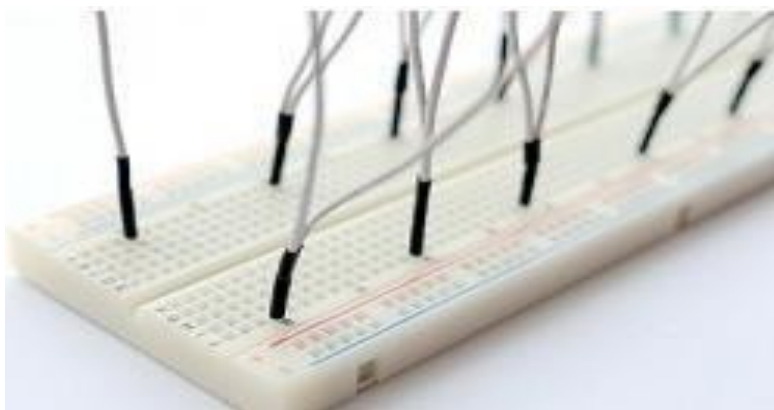
8.2 ESP8266 (WiFi Module)

The ESP8266 is an easy Wi-Fi chip with full TCP/IP stack and MCU (Micro Controller Unit) capacity created by Chinese. These are the primary arrangement of modules made with the ESP8266 by the outsider producer AI-Thinker and remain the most generally available. They are large alluded to as "ESP-xx modules". To shape a workable advancement framework they require extra parts, particularly a serial TTL-to-USB connector and an outside 3.3 volt control supply.



8.3 Bread board and Jump wires

A breadboard is utilized to build and test circuits expeditiously afore finalizing any circuit design. The breadboard has many apertures into which components like ICs and resistors can be connected. The apertures are generally spaced 0.1" apart to put up standard DIP machinery. A typical breadboard that includes top and bottom power distribution rails. Jump wires are generally used to establish connectivity with bread board.



8.4 Power Supply Board

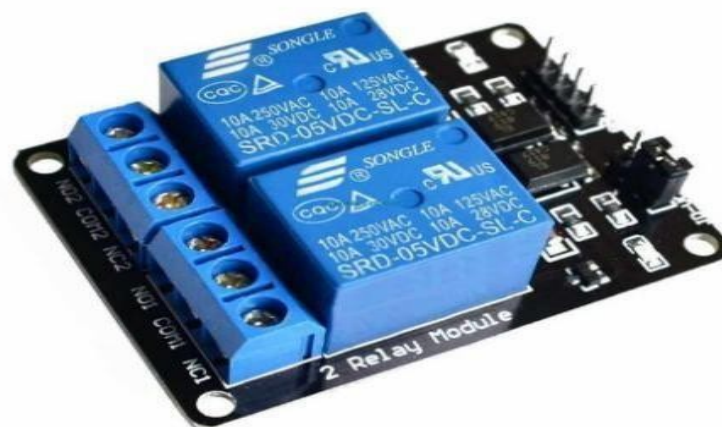
Modern Arduino Uno boards allow the board to have more than one source of power to be connected simultaneously. An intelligent switching circuitry ensures that the highest available voltage is selected and sent to the onboard voltage regulator and eventually powers up the board



8.5 Channel Relay

A relay is an electrically operated switch and is frequently used in automatic control circuit. Relays are used where it is necessary to control a circuit by a low-power signal. Relays protect electrical circuits from overload or faults.

This module is designed to be integrated with 2 relays that it is capable of control 2 relays. The relay shield use one QIANJI JQC-3F high-quality relay with rated load 7A/240VAC, 10A/125VAC, 10A/28VDC. The relay output state is individually indicated by a light-emitting diode.



INTRODUCTION TO IOT PLATFORM

Internet of Things is a multiple layer platform that executes direct management, automation, and provisioning of connected devices within the internet's circumference. It is operated with the help of the cloud, security, and expert systems connected to hardware devices. It is a ready-made platform for beginners and developers for instant use, which operates at a great speed. It is a remote device that allows the user to manage the connection between hardware and application devices. It has a good scope on business development with unique features like cloud-enabled platforms, and IoT enabled the platform.

ARCHITECTURE OF IOT PLATFORM

The architecture comprises four layers: things, connectivity, core IoT attributes, analytics, and applications. The topmost layer of the IoT stack is used for developing connected devices and smart things. In the bottom layer, it has elements for container management, platform management, and IoT orchestration. The communication layer is used for messaging to connected devices. The core layer is used for configuration management, OTA services, and messaging. The top of the core IoT layer is the analytical rule engine and security systems that detect user IoT solutions' anomalies. It is flexible but operated only at a high cost.

TOP PLATFORMS OF IOT

The aim of the Internet of Things is to establish a connection with another IoT device to share information using Internet transfer protocols.

Some of the platform available in the market for building IoT applications are listed below:

1. Microsoft Azure IoT

It is one of Microsoft's topmost platforms and products, which is made to deliver scalable service, application services, messaging facility, data storage services, and efficient databases. The aim of Azure IoT is to give persistent service by efficient data storage and helps the company to give an instant and accurate solution with available data in a repository in every unit of an enterprise.

2. Amazon Web Services

It enables the users with hosting and management of services in cyberspace. Some of the esteemed organizations use AWS to build, host, manage and organize the infrastructure of the company. Here IoT platform offers many benefits to users cloud-based data storage, transfer of application, and manage storage.

3. Google Cloud Platform

It is a popular IoT platform that provides a document-aided database, cloud computing, and elastic database. It is the third-largest IoT platform that powers Machine Intellect, Google analytics, etc.

4. _____ ThingWorx

It is a widely used platform and is acquired by PTC in 2013. The application developers use this platform for easy and effective execution, analysing complicated data and content, and offering solutions to rapid data management.

5. Cisco IoT Cloud Connect

It provides services with effective data and voice connectivity, a reliable SIM lifecycle, proficient IP session control, and a customizable billing system.

6. HP's Universal of Things

It provides accurate solutions to all IT-related issues. It is effectively used for monetization, effective collection of data, data analysis, and the correct platform for launching new mobile applications.

7. _____ SAP Cloud Platform

It is effectively applicable for distantly located devices that are connected in a dual way or direct cloud services. It is equipment with components to build and manage the IoT application.

8. Oracle Internet of Things

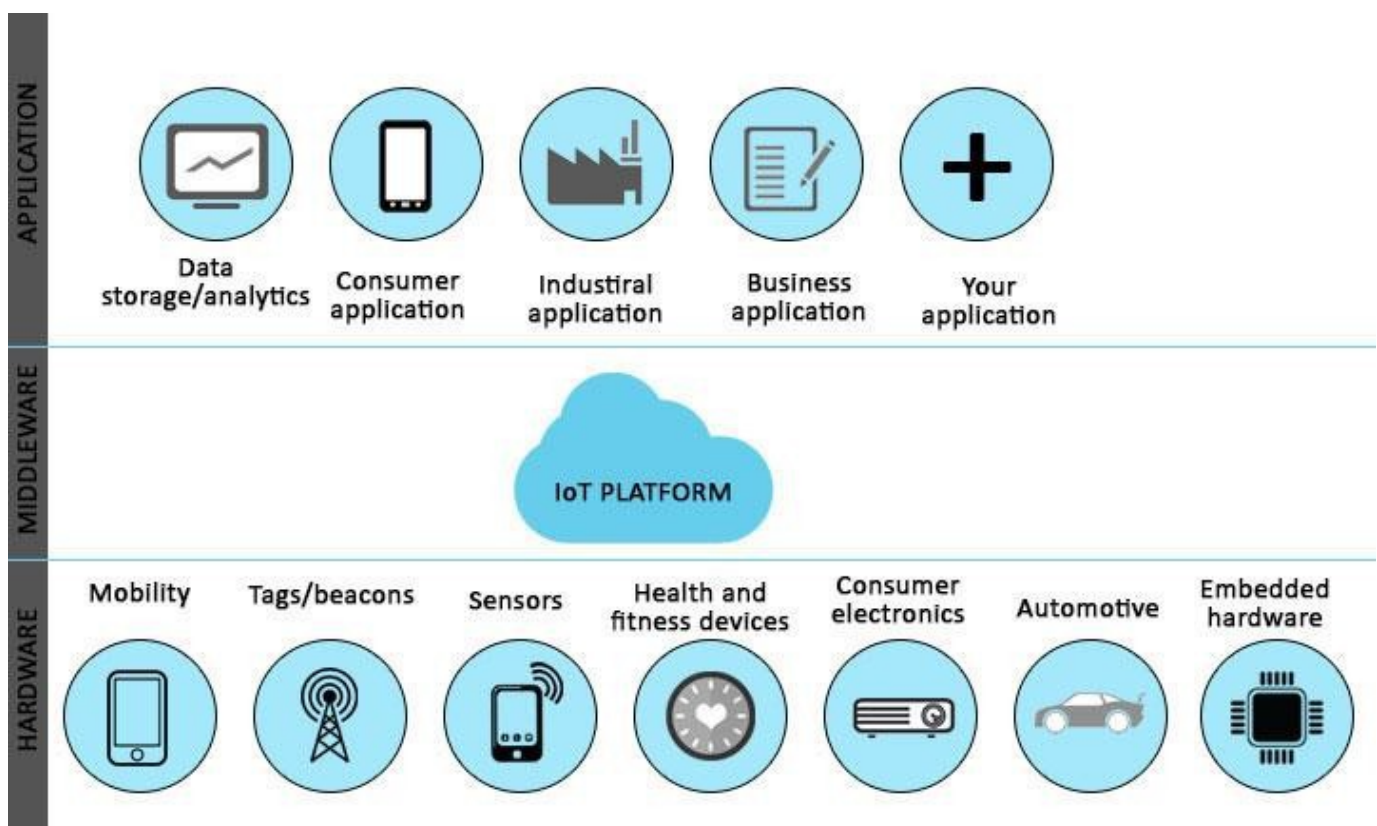
It establishes the connection between a company's software and its policies. It is a flexible environment for developing a new application and use it for commercial purposes.

9. Bosch IoT Suite

It is developed in Germany and stands for its reliable solutions and innovative approach. It is available in an open-source environment.

10. IBM Watson Internet of Things

It offers services like control of remote devices and data transmission without any noises and cloud storage. It offers multiple choices using the IoT platform for Machine Learning and Artificial Intelligence.



INTRODUCTION TO IOT COMPANIES

Internet of things (IoT) has evolved to be the topmost growth driver for business and is highly impacting homes, organizations, and environments across the globe. Internet of Vehicles and smart home technology is already in production and a huge hit. Many companies around the globe are opting for IoT solutions to boost the productivity and efficiency of the company. As a matter of fact, many companies around the globe have already opted for IoT solutions for their business. For example, the banking sector has already invested in IoT solutions to provide rapid service to the customer with the help of kiosks. The health care industry has been incorporating IoT solutions in wearable devices to accumulate patient data.

Examples of IoT Companies

Let us see real-time examples of the companies that have been experimenting with the IoT ecosystem:

1. Alibaba

Alibaba is one biggest E-commerce companies in the market today. This Chinese company has been utilizing IoT to accelerate growth in IoT technology in China. Alibaba officially launched the link IoT platform in the year 2017 to connect billions of devices. Alibaba's cloud link has launched many cloud-based businesses. Alibaba's cloud link has helped numerous businesses across China and the world to incorporate IoT into their ecosystem. A large number of solutions are offered with multiple configuration options. Consumers can configure their hardware and software with ease and not worrying about the cost as they come at very cost-effective prices.

2. DHL

DHL is a Germany based logistics company that has been experimenting with IoT since 2017. With the help of Chinese company Huawei, DHL launched a handbarrow IoT application that allows data transmission over IoT devices at a much lower price. This technology came into existence in order to manage inbound and outbound logistics operations. With the implementation of IoT in logistic services, DHL has successfully monitored the services and activities in real-time.

3. Hitachi

Hitachi is well known for its smart televisions and has launched a new entity called Vantara, which triggered the release of an IoT platform called Lumada for the purpose of turning data into useful resources. With the help of Lumada, an organization can manage and maintain the IoT life cycle of an organization. Lumada video insight is capable of collecting, storing and analyzing video data. An organization can get useful insights and analysis out of data with the help of Lumada tools to maximize efficiency and security.

4. Tesla

Tesla is an American automobile company that has invested heavily in IoT. The company's main motive was to make the car more sustainable and environment-friendly. Every car sold by Tesla has the automation feature, which can add new and latest features to the car with the help of updates rolled out by the company; every Tesla car has the potential to become self-drivable. Tesla has been installing many sensors on the road to collect data, which will help their automobiles be aware of their surroundings. With these features' automobiles are not a fixed commodity but a constantly changing and customizable commodity that can be upgraded with the help of a paid software update. The idea of interconnected cars has been made a reality by Tesla automaker. With advancements in IoT, traditional devices will no longer be fixed commodities but will be able to upgrade themselves and behave smartly.



IoT Companies

HITACHI
Inspire the Next


Alibaba.com
Global trade starts here."

www.educba.com

9. Blynk

Blynk is an **Internet-of-Things** platform designed to make development and implementation of smart IoT devices quick and easy. It can be used to read, store, and visualize sensor data and control hardware remotely.

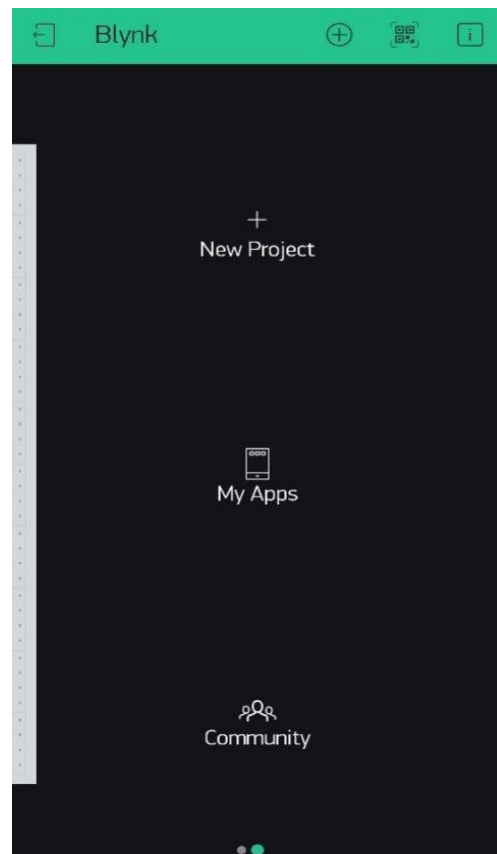
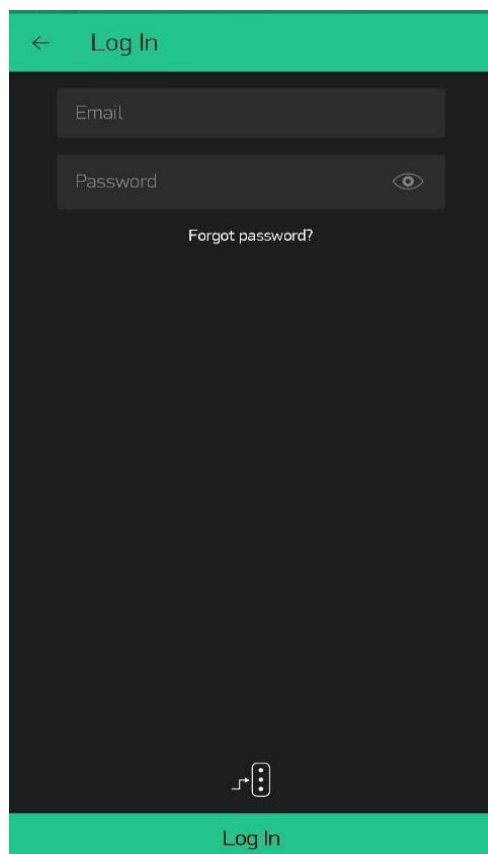
Internet of Things has been all the buzz lately and more and more devices are being talking to internet every day. With the rise of such amazing technology, the risk of security has also increased substantially. Some of the major concerns in IoT are:

- If IoT devices are sending your data to the internet, the communication needs to be closed and encrypted which cannot be possible without using a dedicated and closed server which is really hard to manage.
 - The IoT devices also need to be responsive and again, that is not possible without a server with low latency and high responsiveness.
 - In IoT, the platform needs to be compatible with many different types of hardware architecture and devices, so that it doesn't restrict its users with single type of hardware with limited capabilities.
-
- **Blynk Server** – The most amazing component of the Blynk Platform which makes it all possible is the Blynk Server. Blynk offers a secure, responsive and centralized cloud service through its server which allows all of this communication between the devices. The Blynk server is also available as open source so you can literally make your own server and make it even more secure with a little tinkering.

 - **Blynk Library** – The key feature of Blynk platform which makes it scalable and amazing, is the Blynk Library. The Blynk Library makes it possible to connect your hardware and get it up and running in a blink. The support for multiple hardware devices including **Arduino**, **ESP8266** and **Raspberry Pi** is included in the library and it also makes it possible to connect with hardware through many different ways of communication like Wi-Fi, Bluetooth, BLE, USB and GSM.

10. CONFIGURING BLYNK

After the user installs the Blynk app on the smartphone, an account has to be created in the app to access its services. The first time the app is opened, it will ask to either sign in or create an account. Create an account and add a new project to get started. Each project has its own authentication code which is used by the code to communicate with that particular model. To interface with our components, we need to add widgets to our model. To add widgets press „+“ to add to the model. The app provides a neat interface to add all the required widgets and setting them up according to the code. The Blynk needs to be running in the background for the user to get real time notifications.



11. ADVANTAGES

The main advantage of —Home Automation through Android mobile is that the —physically challenged and disabled people.

- Saves electricity
- Quick response
- Can control home appliances from anywhere from the world
- Helps for safety purpose
- If needed we can be notified by a message
- Monitoring 24/7
- Improved citizen's quality of life
- New business opportunities
- Economical growth
- Better environment
- Improved competitiveness
- Billions of dollars in savings and new services
- IoT can be used in every vertical for improving the efficiency Creates new businesses, and new and better jobs
- Better safety, security and productivity
- Saves natural resources and trees
- Helps in creating a smart, greener and sustainable planet
- Competitive in providing cutting edge products/services

12.DISADVANTAGES

- **Security:** As the IoT systems are interconnected and communicate over networks. The system offers little control despite any security measures, and it can be lead the various kinds of network attacks.
- **Privacy:** Even without the active participation on the user, the IoT system provides substantial personal data in maximum detail.
- **Complexity:** The designing, developing, and maintaining and enabling the large technology to IoT system is quite complicate

13.REQUIREMENTS

10.1 Software Requirements :

1. Arduino IDE

You will be needing Arduino IDE software to write and upload the programming logic onto the Arduino Uno board

2. Blynk App

Also, you need to create an account in the Blynk app IoT to integrate the working of the system physically on the Blynk app.

10.2 Hardware Requirements :

- 1) Power Supply Board
- 2) 2 Channel Relay
- 3) Node MCU (ESP8266)

14. PROJECT IMPLEMENTATION

An application is installed in the mobile (Blynk).

- 1.ESP8266 Wi-Fi module used for providing communication between the application and system.
- 2.By using this we can switch on or off the lights or fans automatically when we are outside
- 3.Also the notification gets triggered in Blynk app whenever a motion is detected

15. SOURCE CODE

```
/* Comment this out to disable prints and save space */
#define BLYNK_PRINT Serial

#include

<ESP8266WiFi.h> #include
<BlynkSimpleEsp8266.h>

// You should get Auth Token in the Blynk App.
// Go to the Project Settings (nut icon).
char auth[] = "2301f2e9572a41aeb096fe4bb31a3fa6";
// Your WiFi credentials.
// Set password to "" for open
networks. char ssid[] =
"Questease Solutions"; char
pass[] = "Fareeha08";

void setup()
{
  // Debug
  console

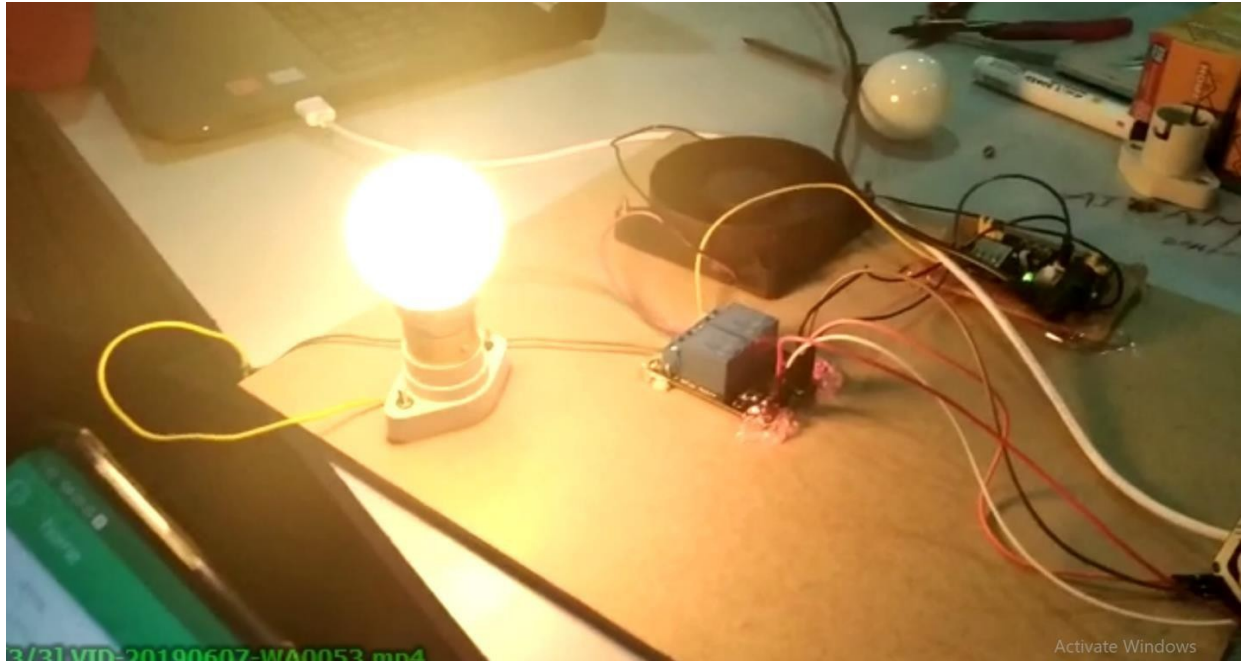
  Serial.begin(9600);
  Blynk.begin(auth, ssid,
pass);

  // You can also specify server:
```

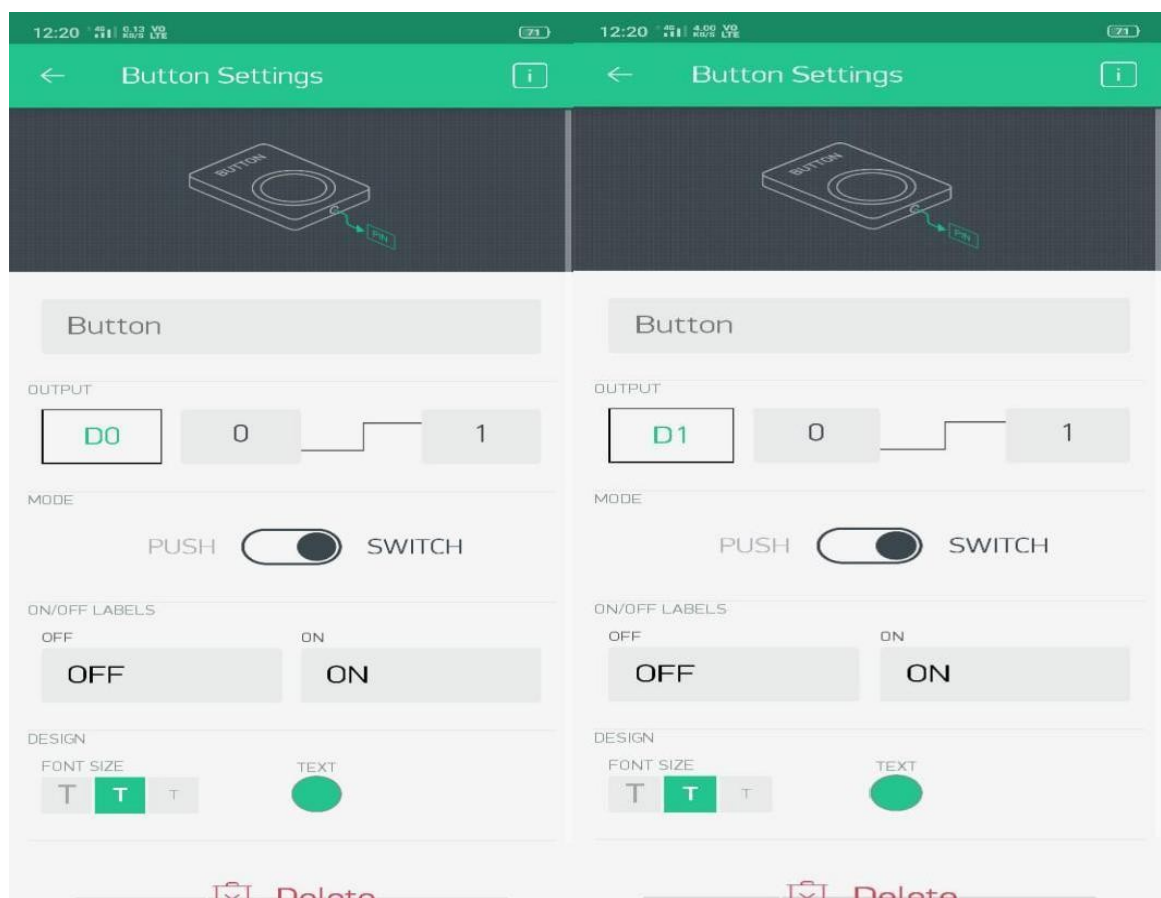
```
//Blynk.begin(auth, ssid, pass, "blynk-cloud.com", 80);  
//Blynk.begin(auth, ssid, pass, IPAddress(192,168,1,100), 8080);  
}  
void loop()
```

```
{  
  Blynk.run();  
}  
void setup() {  
  // put your setup code here, to run  
  once: pinMode(D4,OUTPUT); .  
  
}  
void loop() {  
  // put your main code here, to run  
  repeatedly: digitalWrite(D4,HIGH);  
  
  delay(5000);  
  digitalWrite(D4,LOW);  
  delay(5000);  
  
}  
  
void setup() {  
  // put your setup code here, to run once:  
  pinMode(5,OUTPUT);  
  pinMode(7,OUTPUT);  
  pinMode(9,OUTPUT);  
  
}  
void loop() {  
  // put your main code here, to run  
  repeatedly: digitalWrite(5,HIGH);  
  
  delay(5000);  
  digitalWrite(5,LOW);  
  delay(5000);  
  digitalWrite(7,HIGH);  
  delay(500);  
  digitalWrite(7,LOW);  
  
}
```

16.CIRCUIT CONNECTIONS



OUTPUT



CONCLUSION

The project —IOT BASED HOME AUTOMATION SYSTEM|| has been successfully designed and tested. Integrating features of all the hardware components used have developed it. Presence of every module has been reasoned out and placed carefully thus contributing to the best working of the unit. Secondly, using highly advanced IC's and with the help of growing technology the project has been successfully implemented.

Embedded systems are emerging as a technology with high potential. In the past decades microprocessor based embedded system ruled the market. The last decade witnessed the revolution of Microcontroller based embedded systems. With regards to therequirements gathered the manual work and the complexity in counting can be achieved with the help of electronic devices.

17.BIBLIOGRAPHY

The following books are referred during the analysis and execution phase of the project

- **THE INTERNET OF THINGS** by Samuel Greengard.
- **GETTING STARTED WITH INTERNET OF THING** by Cuno Pfister

REFERENCES :

List of Sample IOT Projects:

- <https://electronicsforu.com/iot-projects-ideas>
- <https://www.skyfilabs.com/blog/best-iot-projects-using-arduino>

Arduino download:

- <https://www.arduino.cc/>

Basic Electronic Components:

- <https://circuitdigest.com/arduino-projects>